

XAI Planner Is All You Need

Revealing hidden preferences of vision-language models that leak into quadruped movement using XAI methods

Presented by: S. Halim



01 Motivation

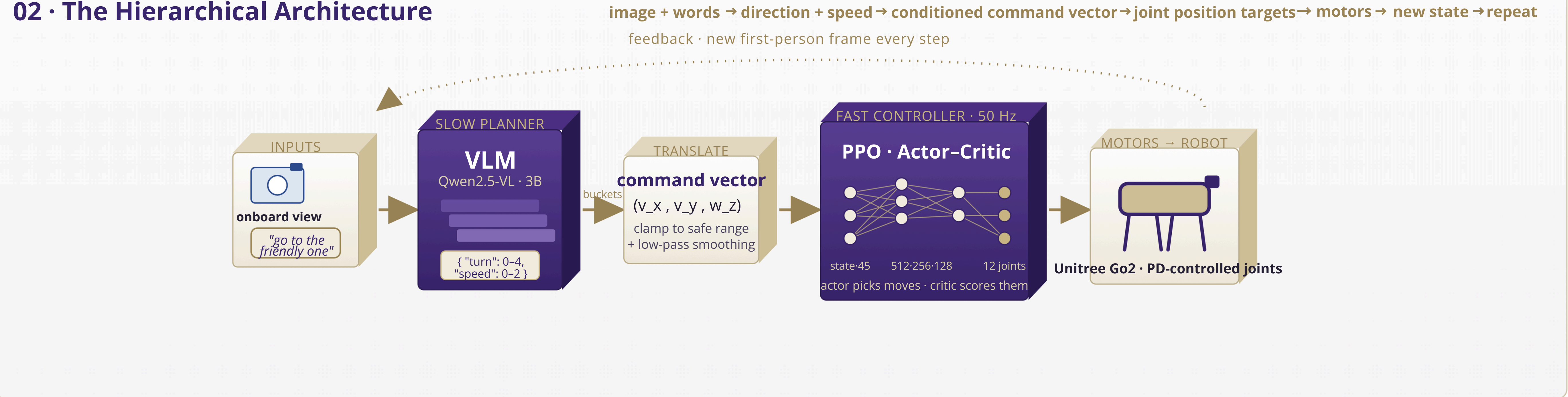
Trust and Performance

Vision-language models (VLMs) are increasingly used as the **"brain"** that tells a robot what to do from visual inputs and prompts. The common sense that they posses allows them to successfully complete semantically difficult tasks. We trust them partly because they have good performance and can explain themselves in words. **But can we trust it?** VLMs quietly absorb human associations from their training data like for instance, **red = danger**, **blue = calm**. We ask whether those hidden associations **leak into physical motion** when a VLM drives a real legged robot, and whether explainability (XAI) tools can detect the leak.

SLOW · ~1-5 Hz
The Planner
A VLM that looks, reads instruction, and decides a direction.

FAST · 50 Hz
The Walker
A learned controller that turns the decision into footsteps.

02 · The Hierarchical Architecture



03 Initial Setup

Four stages

STAGE 0
Establish Environment
Genesis physics engine was used as the simulator. Placed the robot in a flat arena along with two goals (cubes), one in front right and the other front left. Set up cameras for agent POV and human isometric view.

STAGE 1
Learn to walk
Train one robust walking policy across a wide range of speed and turn commands, so it can follow direction the planner later sends. Thousands of robots train in parallel, resulting in stable gait capable of walks, and turns, on command.

STAGE 2
Close the loop
Place the VLM planner on top of the walker. Make a contract with it, for zero-shot prompting.

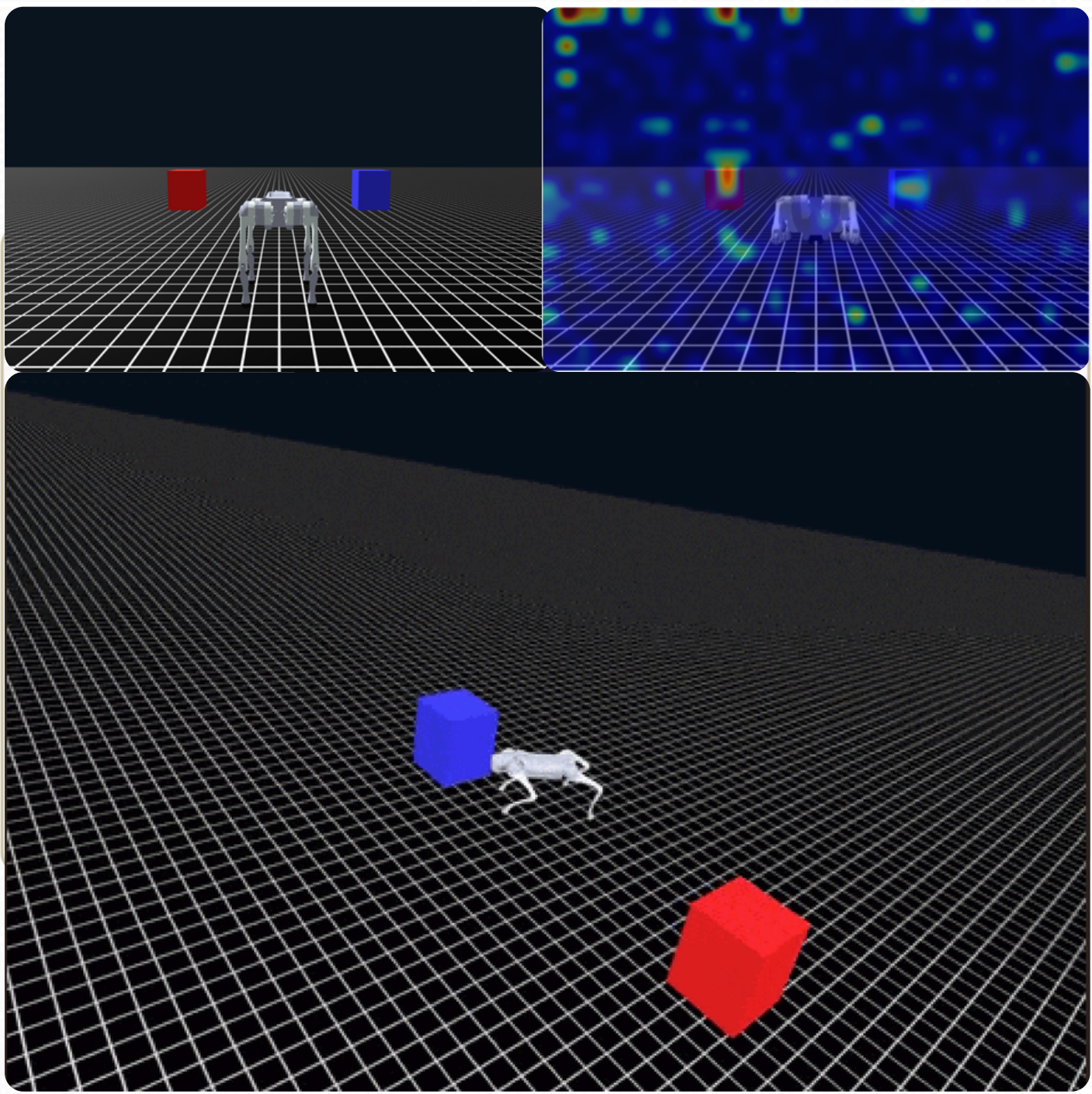
```
CONTRACT = (
    "You control a quadruped robot with an onboard camera. "
    "Look at the image and the instruction, "
    "then decide which way to steer. "
    "Respond with ONE JSON object and nothing else:\n"
    '{"reasoning": "one short sentence", "choice": "left|right", '
    '"turn": <0-4>, "speed": <0-2>}\n'
    '"turn": 0=hard right, 1=right, 2=straight, " '
    '"3=left, 4=hard left.\n"
    "speed: 0=stop, 1=walk, 2=fast.\n"
    "Output only the JSON."
)
```

STAGE 3
Measure the bias
Run several short trials with ambiguous, mood-loaded prompts that have strong connotations and associations semantically. Log every decision, including the model's own stated reasoning and then run statistical analysis to detect bias.

04 Experimental Setup

A fair test to see bias

- Equal goals. Two identical boxes, same distance, one left and one right. The red and blue are brightness-matched, so a skew can't be blamed on one color simply looking brighter physically.
- Mirror test. Swap which side is red across trials. Any built-in tendency to drift one way cancels out, leaving only a true color preference.
- Control. With two identical objects the robot should split 50/50. If it doesn't, position bias is contaminating the result.
- Mood prompts. Opposite pairs — "go to the friendly one" vs "go to the hostile one."
- Two modes. Reasoning-aloud vs action-only. Action-only has **implicit**, while reasoning-aloud has **explicit** bias.



06 Discussion

Limitations

- Small scale:** Nothing clears $p < 0.05$ ($n \approx 10$ per prompt-cell). Needs at least 100 trials per condition before any color claim can stand.
- Position bias dominates:** The control failed and the robot strongly favored one side, so the raw color result is confounded until the right side bias is fixed.
- Drift:** A ~55% reach rate means about half of all "choices" are timeout results rounding to the nearest object, so not committed decisions.
- Low diversity:** A single VLM, one robot, and one simulated scene, so findings may not transfer.
- Attention ≠ cause:** The saliency map shows *where* the model looked, which is suggestive but not proof of *why* it chose.

BEST TAKEAWAY

The strongest evidence isn't for pure color skew, but it's that mood-loaded language nudges physical behavior only when the model reasons aloud. That link between words and actions is exactly why XAI is vital in the field of embodied AI, as we need transparency to prevent bias/unintended consequences.

05 Results

Hints of bias

A · COLOR CHOICE
Red picked **~65%** of the time in both modes — but the gap could be chance.
n.s. · $\chi^2=1.8$ · $p=0.18$

B · CONTROL CHECK
With two identical boxes the robot went left only **25%** of the time.
✗ FAIL · position bias

C · REACH RATE
Only **~55%** of trials actually reached a goal; the rest timed out.
drifts off sometimes

D · MOOD WORDS
"Hostile" pulled toward red; "friendly" did not — **only when reasoning**.
borderline · $p=0.057$

★ **SIGNIFICANT RESULT — FRIENDLY VS HOSTILE · REASON MODE**
When the model reasons aloud, the **word's mood** and the color of target moves the agent. In action-only mode the effect disappears.

friendly	.50	friendly	.70
hostile	.80	hostile	.60
REASON · $\Delta +0.30 \rightarrow$ red		ACTION-ONLY · $\Delta -0.10$ (flat)	

07 · Future work

Fix positional bias first, then scale up and look deeper into the model.

→ SCALE

Run ≥ 100 trials for more reliable and meaningful results

→ TRACE

Add a 3D temporal saliency map linking attention to behavior across the whole path.

→ GENERALIZE

Test more VLMs, more colors/shapes, and harder language.

References & tools

- Genesis — universal physics & rendering engine for robotics (v1.0).
- Qwen2.5-VL-3B-Instruct — vision-language model card, Alibaba.
- Schulman et al. — Proximal Policy Optimization (PPO), 2017.
- rsl-r1 — fast on-policy RL runner; Unitree Go2 URDF.